

HOMEWORK 10 FOR MATH 2860 (ELEMENTARY DIFFERENTIAL EQUATIONS)

E. A. ATINDAMA, PHD MATHEMATICS

Due at 10:00 EST in class

1. Solve the IVP (second-order, sinusoidal forcing):

$$y'' + y = \cos t, \quad y(0) = 0, \quad y'(0) = 0,$$

using Laplace transforms.

Solution

2. Solve the IVP:

$$y'' + y = \mathcal{U}(t - \pi), \quad y(0) = 0, \quad y'(0) = 0.$$

Solution

3. Solve the IVP:

$$y'' + y' = \mathcal{U}(t - 2), \quad y(0) = 0, \quad y'(0) = 1.$$

Solution

4. **Solve** the IVP and **sketch the solution** of

$$y''(t) + 4y(t) = g(t), \quad y(0) = y'(0) = 0$$

for

- (a) $g(t) = \delta(t)$
- (b) $g(t) = \mathcal{U}(t)$
- (c) $g(t) = \delta(t - 1)$
- (d) $g(t) = \mathcal{U}(t - 3)$

Solution